Lab 0 — Inventory your own machine
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```
exlibris@rockstar ~> cat /proc/cpuinfo | grep 'model name' | head -n 1
model name      : AMD Ryzen 7 5800H with Radeon Graphics
exlibris@rockstar ~> nproc
16
exlibris@rockstar ~> grep 'core id' /proc/cpuinfo | sort -u | wc -l
8
exlibris@rockstar ~>
```

1)output the CPU model
2)output core count
3)output thread count.

```
exlibris@rockstar ~> sudo dmidecode --type 17 | awk '/Size:/ {size=$2} /Speed:/ {speed=$2" "$3; if(size !~ "No") print "Модуль: " size ", Скорость: " speed}'
Модуль: 8, Скорость: 3200 MT/s
Модуль: 8, Скорость: Memory Speed:
Модуль: 8, Скорость: 3200 MT/s
Модуль: 8, Скорость: Memory Speed:
exlibris@rockstar ~> free -h
               total        used        free      shared  buff/cache   available
Mem:            14Gi       8,5Gi       3,8Gi       160Mi       3,1Gi       6,4Gi
Swap:           15Gi          0B        15Gi
exlibris@rockstar ~>
```

1) output the number of installed modules and their speed.
2) output find the total memory

```
exlibris@rockstar ~> lsblk -o NAME,MODEL,SIZE,TYPE,FSTYPE,MOUNTPOINTS
NAME        MODEL                          SIZE TYPE FSTYPE MOUNTPOINTS
nvme0n1     SAMSUNG MZVLB512HBJQ-000H1 476,9G disk
├─nvme0n1p1                                 2G part vfat   /boot
├─nvme0n1p2                             458,9G part ext4   /
└─nvme0n1p3                                16G part swap   [SWAP]
exlibris@rockstar ~> lsblk -d -o NAME,MODEL,ROTA,TRAN,SIZE
NAME    MODEL                       ROTA TRAN     SIZE
nvme0n1 SAMSUNG MZVLB512HBJQ-000H1    0 nvme   476,9G
exlibris@rockstar ~> df -h
Файловая система Размер Использовано  Дост Использовано% Смонтировано в
/dev/nvme0n1p2     451G         292G  137G           69% /
devtmpfs           7,4G            0  7,4G            0% /dev
tmpfs              7,5G          79M  7,5G            2% /dev/shm
efivarfs           128K          13K  111K           10% /sys/firmware/efi/efivars
tmpfs              3,0G         1,9M  3,0G            1% /run
none               1,0M            0  1,0M            0% /run/credentials/systemd-journald.service
/dev/nvme0n1p1     2,0G         181M  1,9G            9% /boot
tmpfs              7,5G          27M  7,5G            1% /tmp
tmpfs              1,5G         248K  1,5G            1% /run/user/1000
exlibris@rockstar ~>
```

1)The disk model
2)Type of the disk
3)Free space on the volume you will store VMs on

```
exlibris@rockstar ~> sudo dmidecode -t bios
[sudo] пароль для exlibris:
# dmidecode 3.7
Getting SMBIOS data from sysfs.
SMBIOS 3.3.0 present.

Handle 0x0000, DMI type 0, 26 bytes
Platform Firmware Information
        Vendor: AMI
        Version: F.12
        Release Date: 12/09/2021
        Address: 0xF0000
        Runtime Size: 64 KiB
        ROM Size: 16 MiB
```

The firmware type (BIOS or UEFI) and its version and date.

```
exlibris@rockstar ~ [0|1]> sudo pacman -S msr-tools
                           sudo virt-host-validate
[sudo] пароль для exlibris:
разрешение зависимостей...
проверка конфликтов...

Пакеты (1) msr-tools-1.3-4

Будет загружено:    0,01 MiB
Будет установлено:  0,04 MiB

:: Приступить к установке? [Y/n] ^Zsudo: virt-host-validate: команда не найдена
fish: Job 1, 'sudo pacman -S msr-tools' has stopped
```

code shows that we don’t have hardware virtualization and starts instaling it.